Cbits

$|00\rangle = \begin{pmatrix}1\\0\end{pmatrix} \otimes \begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1\\0\\0\\0\end{pmatrix}$ Tensor/outer product: $V \otimes W = VW^T$

You start by saying you have a binary string of length K

So you have 2^K possible things each $|0\cdots1\cdots\rangle$ is a basis in a space $\mathbb{R}^{2^K}$

- Reversible means for $Ax=b$ given A, b, you can uniquely find x
- Quantum Computers only use reversible operations

CNOT: On pairs of bits, one is "control" bit, other is target bit

$00 \to 00$
$01 \to 01$
$10 \to 11$
$11 \to 10$

$C = \begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&0&1\\0&0&1&0\end{pmatrix}$

$|\psi\rangle$: Vector
$\langle\psi|$: Hermetian of vector
$\langle\psi|\phi\rangle$: Inner product
$|\psi\rangle\langle\phi|$: Outer product

Qbits are represented by $\begin{pmatrix}a\\b\end{pmatrix}$ $a, b \in \mathbb{C}$ $\|a\|^2 + \|b\|^2 = 1$

- Cbit special example

You could say $|\phi\rangle = a|0\rangle + b|1\rangle$, $|0\rangle = \begin{bmatrix}1\\0\end{bmatrix}$, $|1\rangle = \begin{bmatrix}0\\1\end{bmatrix}$

must $\|a\|^2 + \|b\|^2 = 1$

- Superposition: When we measure qbit (end of computation), collapses to either 0 or 1
  - Collapse to 0 with prob $\|a\|^2$ & 1 with prob $\|b\|^2$

- Use Matrices to operate on Qbits:
  - Hardamard gate: $\begin{pmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}}\end{pmatrix}$ $H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$ $H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$

Splits probability equally

So $H|\phi\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1&1\\1&-1\end{bmatrix}\begin{bmatrix}a\\b\end{bmatrix} = \frac{a+b}{\sqrt{2}}|0\rangle + \frac{a-b}{\sqrt{2}}|1\rangle$

$\boxed{\text{Grover's Alg}}$

$H^{\otimes n} = H \otimes H \cdots$ n times

$|0^n\rangle = |0\rangle^{\otimes n} = |0\rangle \otimes |0\rangle \cdots$

- $|0^1\rangle = |0\rangle$
  $|0^2\rangle = |00\rangle$

$H^{\otimes n}|0^n\rangle \overset{n=2}{=} (H|0\rangle) \otimes (H|0\rangle)$

$\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes (|1\rangle + |0\rangle)$

$= \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$

= In general $\frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle = |s\rangle$

(Represent all 2^n bit strings at once.

Say $|w\rangle$ is the correct state

Say $|w^\perp\rangle$ are superpos of incorrect states

$|w^\perp\rangle = \frac{1}{\sqrt{N-1}} \sum_{x \neq w} |x\rangle$ $|s\rangle = \sin(\theta)|w\rangle + \underbrace{\cos(\theta)|w^\perp\rangle}$

$\sin\theta = \frac{1}{\sqrt{N}}$

for each component its $\frac{\cos\theta}{\sqrt{N-1}}$

$\langle\psi(\theta)|H|\psi(\theta)\rangle$

expectation value of operator H on $\psi(\theta)$

Repeat $\sqrt{N}$ times

you tryna get to $\frac{\pi}{2}$ angle, from small θ

2. $U_w|x\rangle$ entire job is to recognize $|w\rangle$

$U_w|x\rangle = \begin{cases} -|x\rangle & x = w \\ |x\rangle & x \neq w \end{cases}$

$U_w = I - 2|w\rangle\langle w|$ reflect across $|w^\perp\rangle$

H is $m \times n$ $[a^* \; b^*]\begin{bmatrix}e&f\\g&h\end{bmatrix}\begin{bmatrix}a\\b\end{bmatrix}$

$[a^* \; b^*]\begin{bmatrix}ea + ga\\ fb + hb\end{bmatrix} =$

3. Diffusion Operator: $2|s\rangle\langle s| - I$ Inversion about the mean

$\alpha_{w^\perp} \to 2\mu - \alpha_{wrong-w^\perp}$ (u get that 3θ change to w axis from initial θ)

$\alpha_w \to 2\mu - \alpha_w$

$(e+g)|a|^2 + (f+h)|b|^2$

$\boxed{= \sum_n \left(\sum_m H^T_{nm}\right) \cdot \psi_n(\theta)}$

Plancks const $\frac{h}{2\pi}$

$\psi: (\vec{x}, t): \mathbb{R}^3 \times \mathbb{R} \to \mathbb{C}$

Wave eq: $A\cos(Kx - \omega t)$

$\cos(x)$ vs $\cos(2x)$ (tighter waves)

$x - \omega t$. Wave peak travels velocity t from x.

Time Dependent Schrodinger: $i\hbar \frac{\partial}{\partial t}\psi(\vec{x}, t) = \hat{H}\psi(\vec{x}, t)$ like $\frac{d\vec{x}}{dt} = A\vec{x}$

$\hat{H} = \frac{-\hbar^2}{2m}\nabla^2 + V(x)$ (one electron)

E = total Energy $= \frac{p^2}{2m} + V$ $p = mv$ $\frac{1}{2}mv^2$

$e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots$

$(e^{-i\theta H})^*(e^{-i\theta H}) = I$ scale, direction H is a Hermetian matrix

$U = e^{-i\theta H}$

(Small change H propagated by e, scaled by $\theta = U$)

2 particles, either fermions or bosons

- Fermions can only occupy one state at a time
  - State can be anything, like momentum, energy, spin etc.
- Bosons can have multiple of the same state

Spin up - $|0\rangle$
Spin down - $|1\rangle$

Quantum Gates

Pauli X gate

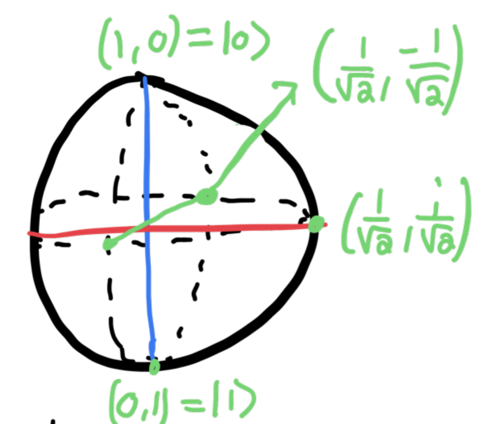
$|1\rangle - \boxed{X} \to |0\rangle$ $X = \begin{bmatrix}0&1\\1&0\end{bmatrix}$

Hadamard: $H = \frac{1}{\sqrt{2}}\begin{bmatrix}1&1\\1&-1\end{bmatrix}$

- These logic gates must be
  - Hermitian: $U = U^*$
  - Unitary: $UU^* = I$

$|0\rangle - \boxed{Y} - i|1\rangle$ $Y|0\rangle = \begin{bmatrix}0&-i\\i&0\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}0\\i\end{bmatrix} \equiv i|1\rangle$

$|+\rangle - \boxed{Z} \to |-\rangle$ $Z|+\rangle = \begin{bmatrix}1&0\\0&-1\end{bmatrix}\begin{pmatrix}1/\sqrt{2}\\1/\sqrt{2}\end{pmatrix} = \begin{bmatrix}1/\sqrt{2}\\-1/\sqrt{2}\end{bmatrix} \equiv |-\rangle$



Bloch Sphere

4 deg $|\psi\rangle = a|0\rangle + b|1\rangle$

2 deg $|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$

Spin $\frac{1}{2}$ rotation operator: $D(\hat{n}, \phi) = \cos\left(\frac{\phi}{2}\right)I - i\sin\left(\frac{\phi}{2}\right)\vec{\sigma} \cdot \hat{n}$

Rotation Gates

Rotation of θ around

x-axis: $R_x(\theta) = \begin{pmatrix}\cos\left(\frac{\theta}{2}\right) & -i\sin\left(\frac{\theta}{2}\right)\\ i\sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right)\end{pmatrix}$

y-axis: $R_y(\theta) = \begin{pmatrix}\cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right)\\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right)\end{pmatrix}$

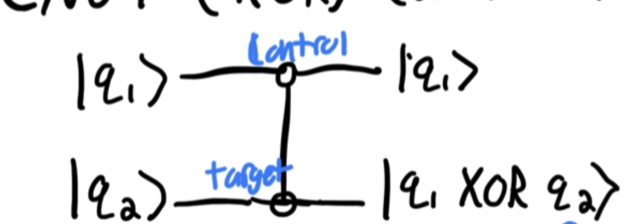
Z-axis: $R_z(\theta) = \begin{pmatrix}e^{\frac{-i\theta}{2}} & 0\\ 0 & e^{\frac{i\theta}{2}}\end{pmatrix}$

For a general n-qubit system

$|\psi\rangle = \sum_{K=1}^{2^n} c_K|\phi_K\rangle$: Superposition

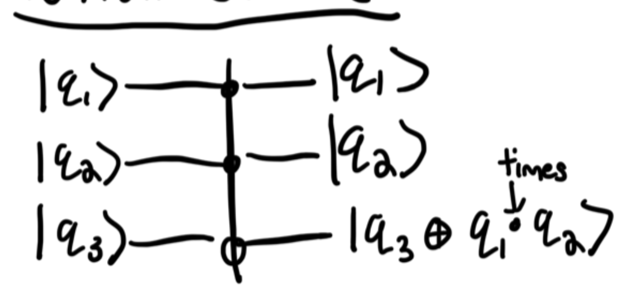
$U = \sum_{j,K=1}^{2^n} U_{jK}|\phi_j\rangle\langle\phi_K|$: Gate operator

CNOT (XOR) (2 Qubits)



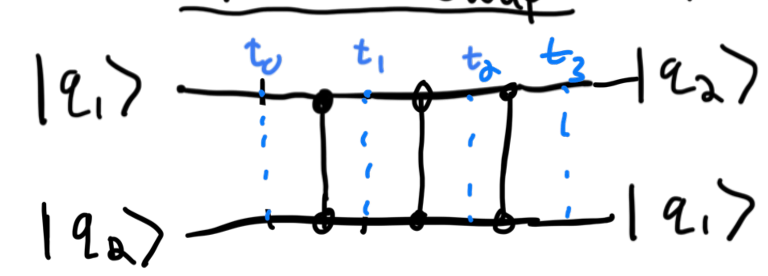
$U_{cnot} = \begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&0&1\\0&0&1&0\end{bmatrix}$

Toffoli Gate (Controlled CNOT)



2 Qubit Swap

A XOR (A XOR B) = B



$t_0: |\langle q_1, q_2\rangle$

$t_1: |q_1, q_1 \text{ XOR } q_2\rangle$

$t_2: |q_1 \oplus (q_1 \oplus q_2), \ldots\rangle$

$t_3: |q_2, q_2 \oplus (q_2 \oplus q_1) = q_1\rangle$